

# Digital Signal Processing — Master Formula Sheet

CSE 4631 · Complete Mathematical & Analytical Quick Reference | LTI Systems, Sampling, DFT/FFT, FIR/IIR Filters, Spectral Analysis & Modulation

## 1. SIGNALS & LTI SYSTEM PROPERTIES

### 1.1 Linearity & Time-Invariance Tests

- **Linearity Test (Superposition Principle):**

$$T\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot T\{x_1[n]\} + b \cdot T\{x_2[n]\}$$

Rule: Non-zero DC offsets, powers (e.g.  $x^2[n]$ ), or non-linear functions break linearity.

- **Time-Invariance Test:**

$$T\{x[n - k]\} = y[n - k]$$

Rule: Any explicit dependence on time index  $n$  (e.g.  $y[n] = n \cdot x[n]$ ) makes the system time-variant.

### 1.2 Energy, Power & Signal Decompositions

- **Total Energy & Average Power Formulas:**

$$\text{Energy: } E = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad | \quad \text{Power: } P = \lim_{N \rightarrow \infty} [1 / (2N + 1)] \sum_{n=-N}^N |x[n]|^2$$

Unit Step Example:  $u[n]$  has Infinite Energy ( $E = \infty$ ) and Finite Power ( $P = 0.5 \text{ W}$ )  $\rightarrow$  Power Signal.

- **Even & Odd Component Decomposition:**

$$x_{\text{even}}[n] = 0.5 \cdot (x[n] + x[-n]) \quad | \quad x_{\text{odd}}[n] = 0.5 \cdot (x[n] - x[-n])$$

- **Interlaced Decomposition:** Splits sequence  $x[n]$  into even-indexed  $x_{\text{even}}[n]$  and odd-indexed  $x_{\text{odd}}[n]$  arrays.

## 2. SAMPLING, NYQUIST, ALIASING & QUANTIZATION

### 2.1 Sampling Theorem & Aliasing Math

- **Nyquist Criterion & Folding Frequency:**

$$f_s > 2 \cdot f_{\text{max}} \Rightarrow \text{Nyquist Rate} = 2 \cdot f_{\text{max}} \quad | \quad \text{Folding Frequency } f_{\text{folding}} = f_s / 2$$

- **Aliased Frequency Formula:**

$$f_{\text{alias}} = |f - k \cdot f_s| \quad (\text{chosen such that } 0 \leq f_{\text{alias}} \leq f_s / 2)$$

Example: 700 Hz tone sampled at  $f_s = 1000 \text{ Hz}$  aliases to  $|700 - 1000| = 300 \text{ Hz}$ .

### 2.2 Quantization & Signal-to-Quantization-Noise Ratio (SQNR)

- **Quantization Resolution & Bit Rate:**

$$L = 2^b \text{ levels} \Rightarrow \Delta = V_{\text{pp}} / L = (V_{\text{max}} - V_{\text{min}}) / 2^b \quad | \quad \text{Bit Rate } R = b \cdot f_s \text{ (bits/s)}$$

- **SQNR Benchmark Equation:**

$$\text{SQNR} \approx 6.02 \cdot b + 1.76 \text{ dB}$$

Rule of Thumb: Every additional ADC bit increases SQNR by approximately 6 dB.

### 3. DISCRETE FOURIER TRANSFORM (DFT) & FOURIER FAMILY

#### 3.1 The Fourier Family & Duality Golden Rule

| Transform Name          | Time Domain            | Frequency Domain       | Application                        |
|-------------------------|------------------------|------------------------|------------------------------------|
| Fourier Transform (FT)  | Continuous & Aperiodic | Continuous & Aperiodic | Continuous analog signal analysis  |
| Fourier Series (FS)     | Continuous & Periodic  | Discrete & Aperiodic   | Periodic analog wave harmonics     |
| Discrete-Time FT (DTFT) | Discrete & Aperiodic   | Continuous & Periodic  | Theoretical discrete equations     |
| Discrete FT (DFT)       | Discrete & Periodic    | Discrete & Periodic    | Digital computers & DSP algorithms |

• **Duality Golden Rule:** Discrete in one domain  $\Leftrightarrow$  Periodic in the other domain. Continuous in one domain  $\Leftrightarrow$  Aperiodic in the other domain.

#### 3.2 Real-DFT Analysis, Synthesis & Conjugate Symmetry Rules

• **Real DFT Analysis Equations (Forward DFT):**

$$\text{ReX}[k] = \sum_{n=0}^{N-1} x[n] \cos(2\pi k n / N) \quad | \quad \text{ImX}[k] = -\sum_{n=0}^{N-1} x[n] \sin(2\pi k n / N)$$

• **Real-DFT Symmetry Rules (Conjugate Symmetry):**

$$\text{ReX}[k] = \text{ReX}[N - k] \text{ (Even)} \quad | \quad \text{ImX}[k] = -\text{ImX}[N - k] \text{ (Odd)} \quad | \quad \text{ImX}[0] = 0 \text{ (DC)} \quad | \quad \text{ImX}[N/2] = 0 \text{ (Nyquist)}$$

*Packaging:* N real time samples  $\rightarrow$  N/2+1 complex bins (N+2 total numbers, exactly N independent values).

• **Inverse DFT Synthesis Equation & Scaling Factors:**

$$x[n] = \sum_{k=0}^{N/2} \text{ReX\_bar}[k] \cos(2\pi k n / N) - \sum_{k=0}^{N/2} \text{ImX\_bar}[k] \sin(2\pi k n / N)$$

*Scaling Rules:* Interior bins ( $1 \leq k \leq N/2 - 1$ ) scale by 2/N. Endpoints ( $k=0$  and  $k=N/2$ ) scale by 1/N.

#### 3.3 Frequency Axis Mapping & Spectral Density

• **4 Frequency Labeling Methods:** Bin index  $k \in [0, N/2] \Rightarrow$  Fraction  $f = k / N \in [0, 0.5] \Rightarrow$  Radians  $\omega = 2\pi k / N \in [0, \pi] \Rightarrow$  Hz  $f_{\text{Hz}} = (k / N) \cdot f_{\text{s}}$ .

• **Spectral Density Bandwidth:** Interior samples represent a bandwidth of 2/N; boundary samples (DC and Nyquist) represent 1/N bandwidth.

### 4. FAST FOURIER TRANSFORM (FFT) & COMPUTATION

#### 4.1 Complexity, Bit-Reversal & Butterfly Unit

• **Computational Complexity:** Direct DFT requires  $O(N^2)$  real operations; Radix-2 FFT requires  $O(N \log_2 N)$  operations.

**Speedup  $S = N^2 / (N \log_2 N) = N / \log_2 N$  (e.g.  $N = 1024 \Rightarrow S = 102.4\times$  faster & more accurate).**

• **Bit-Reversal Indexing:** Index n written in  $\log_2 N$  bits, reversed bit-order gives target position.

*Example ( $N=16$ ):*  $n = 3$  (0011\_2)  $\rightarrow$  12 (1100\_2) |  $n = 7$  (0111\_2)  $\rightarrow$  14 (1110\_2).

• **FFT Butterfly Calculation Element:**

$$A = X_1 + W_N^k X_2 \quad | \quad B = X_1 - W_N^k X_2 \text{ where Twiddle Factor } W_N^k = e^{-j 2\pi k / N}$$

• **Spectrum Combination:**  $X(k) = E(k) + W_N^k O(k)$  |  $X(k + N/2) = E(k) - W_N^k O(k)$  for  $k = 0 \dots N/2 - 1$ .

#### 4.2 Segment Averaging (Welch's Method)

• **Spectral Noise Reduction Law:**

$$\text{Noise Amplitude} \propto 1 / \sqrt{N_{\text{segments}}}$$

*Rule:* Averaging magnitude spectra of 100 windowed segments reduces noise amplitude by a factor of 10 (10 dB) while preserving frequency resolution.

## 5. POLAR NOTATION & THE 7 NUISANCES

### 5.1 Rectangular-to-Polar Conversions & Quadrant Corrections

- **Rectangular-to-Polar Formulas:**

$$\text{MagX}[k] = \sqrt{(\text{ReX}[k]^2 + \text{ImX}[k]^2)} \quad | \quad \text{PhaseX}[k] = \arctan(\text{ImX}[k] / \text{ReX}[k]) + \text{Correction}$$

- **Quadrant Correction Rules:**

If  $\text{ReX} < 0$  and  $\text{ImX} < 0$ : Subtract  $\pi$  ( $-180^\circ$ ). If  $\text{ReX} < 0$  and  $\text{ImX} \geq 0$ : Add  $\pi$  ( $+180^\circ$ ).

### 5.2 Summary of the 7 Polar Nuisances in DSP

1. **Radians vs. Degrees:** Trig functions require radians; define  $\text{PI} = 3.14159265$ .
2. **Divide by Zero:** Occurs when  $\text{ReX} = 0$ ; substitute  $\text{ReX} = 1\text{e-}20$  or set phase to  $\pm\pi/2$ .
3. **Arctan Quadrant Ambiguity:** Standard arctan yields  $[-\pi/2, \pi/2]$ ; requires  $\pm\pi$  offset when  $\text{ReX} < 0$ .
4. **Phase Instability at Small Magnitudes:** When  $\text{MagX} \approx 0$ , round-off noise causes wild phase swings.
5.  **$2\pi$  Ambiguity & Phase Unwrapping:** Discontinuities at  $\pm\pi$  fixed by adding/subtracting integer multiples of  $2\pi$ .
6. **Magnitude Positivity ( $\pi$  Ambiguity):** Positive magnitude constraint forces  $\pi$  phase jumps on negative real dips.
7. **Boundary Spikes:** Round-off noise near  $\pm\pi$  causes rapid toggling between  $+\pi$  and  $-\pi$ .

## 6. DIGITAL FILTER DESIGN & KERNEL TRANSFORMATIONS

### 6.1 Windowed-Sinc Low-Pass FIR Filter Design

- **Unwindowed Ideal Sinc Kernel (Shifted by  $M/2$ ):**

$$h[i] = K \cdot \sin(2\pi f_c (i - M/2)) / [\pi (i - M/2)] \quad \text{for } i \neq M/2 \quad | \quad h[M/2] = K \cdot 2\pi f_c$$

- **Kernel Length (M) & Transition Bandwidth (BW):**

$$M \approx 4 / \text{BW} \quad (\text{where } \text{BW} = \text{transition\_width} / f_s, f_c = \text{cutoff} / f_s)$$

- **Window Constants & Attenuation Specs:**

**Rectangular:**  $M \approx 0.9 / \text{BW}$  (-21 dB stopband atten) | **Hamming:**  $M \approx 3.3 / \text{BW}$  (-53 dB atten, 0.2% ripple) | **Blackman:**  $M \approx 5.5 / \text{BW}$  (-74 dB atten, 0.02% ripple).

- **Window Functions ( $0 \leq i \leq M$ ):**

$$\text{Hamming: } w[i] = 0.54 - 0.46 \cos(2\pi i / M) \quad | \quad \text{Blackman: } w[i] = 0.42 - 0.5 \cos(2\pi i / M) + 0.08 \cos(4\pi i / M)$$

- **Unity Gain Normalization:**  $h_{\text{norm}}[i] = h[i] / \sum_j h[j]$  (guarantees gain = 1 at DC).

### 6.2 Filter Transformations & Multi-Stage Cascading

- **Spectral Inversion (LP  $\rightarrow$  HP, Same Cutoff):**

$$h_{\text{HP}}[i] = \delta[i - M/2] - h_{\text{LP}}[i] \quad (\text{Negate all samples, then add } +1 \text{ at center index } M/2)$$

- **Spectral Reversal (LP  $\rightarrow$  HP, Flips Spectrum about  $f_s/4$ ):**

$$h_{\text{HP}}[i] = (-1)^i \cdot h_{\text{LP}}[i] \quad (\text{Modulates by } 0.5 f_s; \text{cutoff becomes } 0.5 - f_c)$$

- **Band-Pass & Band-Reject Combination Rules:**

$$\text{Band-Pass: } h_{\text{BP}} = h_{\text{HP}}(\text{low}) * h_{\text{LP}}(\text{high}) \quad (\text{Cascade/Convolve}) \quad | \quad \text{Band-Reject: } h_{\text{BR}} = h_{\text{LP}}(\text{low}) + h_{\text{HP}}(\text{high}) \quad (\text{Parallel/Add})$$

- **Cascaded Filter Self-Convolution:**  $h_{\text{comb}} = h * h$  roughly doubles stopband attenuation in dB (e.g. -74 dB  $\rightarrow$  -148 dB).

## 7. CONVOLUTION, DUALITY, MODULATION & DECIBELS

### 7.1 Linear Convolution, Circular Wrap & Time-Shift Phase

- **Discrete Linear Convolution & Output Length:**

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \Rightarrow \text{Output Length } L = N + M - 1$$

- **Avoiding Circular Convolution:** Zero-pad time-domain sequences so DFT length  $L \geq N + M - 1$ .

- **Time Shift Phase Property:**

$$\Delta\text{Phase}[k] = -2\pi k s / N = -2\pi f s \text{ radians (for time delay of } s \text{ samples)}$$

*Key Points:* Phase shift is 0 at DC ( $f=0$ ), linear across frequencies, and exactly  $-\pi$  rad ( $-180^\circ$ ) at Nyquist ( $f=0.5$ ) for a 1-sample shift.

### 7.2 Amplitude Modulation & Decibel Math

- **Amplitude Modulation Spectrum Shift:**

$$y[n] = x[n] \cdot \cos(\omega_c n) \Leftrightarrow Y(\omega) = 0.5 X(\omega - \omega_c) + 0.5 X(\omega + \omega_c)$$

- **Decibel Calculations (Power vs. Amplitude):**

$$\text{dB\_power} = 10 \log_{10}(P_2 / P_1) \mid \text{dB\_amplitude} = 20 \log_{10}(A_2 / A_1)$$

*Key Benchmarks:* -3 dB  $\rightarrow$  Amplitude = 0.707 (Power = 0.5) | -20 dB  $\rightarrow$  Amplitude = 0.1 (Power = 0.01).

- **Absolute Reference Units:** dBV (referenced to 1V rms), dBm (referenced to 1mW into 600  $\Omega$  load  $\approx$  0.78V rms).